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Journal of Alloys and Compounds 334 (2002) 173–181

Journal of ALLOYS AND COMPOUNDS

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Thermodynamic optimization of the B–Fe system

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Received 2 July 2001; accepted 18 July 2001

Abstract

The B–Fe binary system has been optimized using the CALPHAD method, utilizing published experimental thermochemical and phase diagram data. Two models for the solid solubility of B in b.c.c. Fe and in f.c.c. Fe are presented, B as a interstitial, and B as a substitutional constituent. Thermodynamic calculations based on the interstitial and the substitutional model were in good agreement with experimental data. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Transition metal alloys; Phase diagram; Thermodynamic modelling

1. Introduction

The knowledge of the thermodynamic properties of the B–Fe system can be of interest in several fields of materials engineering: e.g. B–Fe is a subsystem of the B–Fe–Nd alloys with superior magnetic properties [1,2], and the high modulus TiB_2-reinforced steel composite is based on the B–Fe–Ti ternary [3]. Furthermore, boron is used as an alloying element to enhance the hardenability of steels and to form amorphous alloys.

Several critical evaluations of the B–Fe system have been reported in literature [4–7]. Thermodynamic calculations of the system were also presented [4,7–10]. Chart [4] proposed a six-parameter model for the liquid and an estimation for the dilute solutions of B in b.c.c. and f.c.c. Fe. Kaufman et al. [8] used a regular solution for the liquid phase, and a subregular model for the b.c.c. and f.c.c. phases. Ohtani et al. [9] considered the interstitial solid solution of B in b.c.c. and f.c.c. Fe to calculate the Fe–B–C ternary. On the other hand, Hallemans et al. [7] suggested substitutional solubility of B in both Fe phases. The latter description and the calculation by Pan [10] used the SGTE recommended expressions for pure Fe and B, given by Dinsdale [11].

In this work a thermodynamic description of the stable phases in the B–Fe system is presented, based on the available experimental data. As compared to the most recent reassessment [7], five additional sources of experimental data were considered [12–16].

2. Experimental data

In the B–Fe system, six equilibrium phases are present, viz. the liquid (L), three terminal solid solutions, b.c.c. Fe (αFe and δFe), f.c.c. Fe (γFe) and rhombohedral B (βB), and two intermediate phases Fe_2B and FeB. The Fe_3B phase reported by Takahashi et al. [13] and by Khan et al. [17] is generally considered to be metastable [18]. The occurrence of the FeB_2 phase, reported by Voroshin et al. [19], was not reproduced by other authors.

For the B–Fe system, only limited phase diagram data are available. (Table 5 presented at the end of the Discussion includes literature data on the invariant equilibria, as reported by several investigators [19–22]). Kneller and Khan [23] suggest that the Fe_2B compound melts congruently. Due to the discrepancy with all other data, this is not used in the optimization.

Experimental phase boundary data are available for the liquidus [20], and for the solid solubility of B in Fe [22,24–30]. In Table 1, these data and the corresponding experimental techniques are presented. In this optimization only the most recent data by Cameron and Morral [22] on the solid solubility of B in Fe are considered. They used the autoradiography technique which allows to discriminate between B in solution, B contained in precipitates, or

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PII: S0925-8388(01)01777-7

Table 1
Experimental phase boundary data for the B–Fe binary system

Phase boundary investigated	Experimental technique used	Ref.
Liquidus	Thermal analysis	[20]
$\alpha Fe/(\alpha Fe+Fe_2B)$, $\gamma Fe/(\gamma Fe+Fe_2B)$	Autoradiography	[22]
$\alpha Fe/(\alpha Fe+Fe_2B)$, $\gamma Fe/(\gamma Fe+Fe_2B)$	Spectrography	[24]
$\alpha Fe/(\alpha Fe+Fe_2B)$, $\gamma Fe/(\gamma Fe+Fe_2B)$, $\alpha Fe/(\alpha Fe+\gamma Fe)$, $\gamma Fe/(\alpha Fe+\gamma Fe)$	Metallography, Thermal analysis, Spectrography	[25]
$\alpha Fe/(\alpha Fe+Fe_2B)$, $\gamma Fe/(\gamma Fe+Fe_2B)$	X-ray diffraction	[26]
$(\alpha Fe+Fe_2B)/(\gamma Fe+Fe_2B)$	Metallography	[27]
$\gamma Fe/(\alpha Fe+\gamma Fe)$	Dilatometry, Internal friction	[28]
$\alpha Fe/(\alpha Fe+Fe_2B)$, $\gamma Fe/(\gamma Fe+Fe_2B)$, $\gamma Fe/(\gamma Fe+Liquid)$	Autoradiography	[29]
$\gamma Fe/(\alpha Fe+\gamma Fe)$, $\alpha Fe/(\alpha Fe+\gamma Fe)$	Resistivity	[30]

B segregated at grain boundaries. In addition, they used high purity starting material. It is generally accepted [15] that the large scatter in the other experimental data is due to the difficulty for most techniques to distinguish between the different appearances of boron in the samples.

Because of the very low solubility of B in all solid solutions of Fe, the invariant reactions concerning these phases, viz. $\delta Fe+L \rightleftharpoons \gamma Fe$ and $\gamma Fe \rightleftharpoons \alpha Fe+Fe_2B$, are nearly degenerated. The invariant temperatures of these reactions are therefore very close to the transition temperature for $\delta Fe \rightleftharpoons \gamma Fe$ (1667.5 K), and for $\gamma Fe \rightleftharpoons \alpha Fe$ (1184.8 K), respectively. The former reaction is generally believed to be of the metatectic type [6,15], while the latter is thought to be an eutectoid reaction, according to the latest experimental information [22,30].

Several authors have reported thermochemical data on the binary B–Fe system [14,21,30–36]. The data used in the optimization are summarized in Table 2. The measurements of the heat capacity of the FeB compound in the temperature range 5–300 K, reported by Sirota et al. [16] could not be taken into account, since the SGTE expressions for the Gibbs free energy are given only above 298 K.

Table 2
Experimental thermochemical data for the B–Fe binary system

Quantity measured	Phase	Experimental technique	Ref.
Enthalpy of formation	FeB	Calorimetry	[21]
Activity of B	αFe, γFe	Photometry	[30]
Enthalpy of formation	Fe_2B, FeB	Calorimetry	[31]
Gibbs energy of formation	Fe_2B, FeB	EMF	[32]
Partial enthalpy of mixing	Liquid	Calorimetry	[33]
Gibbs energy of formation	Fe_2B, FeB	EMF	[34]
Enthalpy of formation	Fe_2B, FeB	Calorimetry	[14]
Enthalpy of formation	Fe_2B, FeB	EMF	[35]
Activity of B	Liquid	Calorimetry	[36]

3. Thermodynamic models

Using appropriate sublattice models [37,38], the Gibbs free energies of the equilibrium phases can be modelled. The thermodynamic descriptions of the pure elements are taken from the SGTE recommended database [11]. In this database the Standard Element Reference (SER) is used as the reference state for Gibbs energies.

The liquid phase was modeled as a single-lattice random solution. The integral molar Gibbs free energy G_m^L is given by:

$$G_m^L = \sum_{i=B,Fe} x_i \, {}^\circ G_i^L + RT \sum_{i=B,Fe} x_i \ln x_i + x_B x_{Fe} L_{B,Fe}^L \quad (1)$$

x_i is the mole fraction of component i in L, and ${}^\circ G_i^L$ is the molar Gibbs free energy of pure liquid i. The excess energy term $L_{B,Fe}^L$ represents the interaction between B and Fe in the liquid L. In order to account for the composition dependence, $L_{B,Fe}^L$ can be expanded as an nth degree Redlich–Kister polynomial [39]:

$$L_{B,Fe}^L = \sum_{\nu=0}^{n} {}^\nu L_{B,Fe}^L (x_B - x_{Fe})^\nu \quad (2)$$

The interaction coefficients ${}^\nu L_{B,Fe}^L$ may have a temperature dependence, i.e.:

$${}^\nu L_{B,Fe}^L = {}^\nu a^L + {}^\nu b^L T. \quad (3)$$

The solid solutions of B in b.c.c. Fe and in f.c.c. Fe were experimentally observed to exhibit a mixed substitutional-interstitial character [28–30,40]. Previous thermodynamic calculations of the B–Fe system used an interstitial model [9], or a substitutional model for the solid solution of B in Fe [7]. In this work, both models are described: B as an interstitial, and as a substitutional constituent. Since the solid solubility of B in Fe is extremely limited, the choice of a particular solution model for B in Fe is not expected to make any quantitative difference. However, consistency is needed when this binary system is extrapolated to higher-order systems. Therefore, both descriptions are necessary, e.g. B is considered as an interstitial species in the B–Fe–Ti ternary [3], but as a substitutional species in the B–Fe–Nd system [2].

In both cases, the b.c.c. and f.c.c. solid solutions are modelled using a two-sublattice model: $(P,\ldots)_p:(Q,\ldots)_q$, where the first sublattice contains the substitutional species, and the second sublattice is for the interstitials (including vacancies □). Therefore, $(Fe)_1:(B,\square)_c$ is used as the interstitial solid solution model, and $(B,Fe)_1:(\square)_c$ in the substitutional case. The index c represents the relative number of sites in the interstitial sublattice and, thus, depends on the crystal structure: 3 for b.c.c. and 1 for f.c.c. The molar Gibbs free energy for phase ϕ is (ϕ = b.c.c. or f.c.c.)

$$^{\circ}G_{m}^{\phi} = \sum_{i=B,\square} {}^{1}y_{Fe}\ {}^{2}y_{i}\ {}^{\circ}G_{Fe:i}^{\phi} + cRT \sum_{i=B,\square}^{2} y_{i} \ln {}^{2}y_{i} + {}^{xs}G_{m}^{\phi} + {}^{mo}G_{m}^{\phi} \quad (4)$$

for the interstitial model $(Fe)_1$:$(B,\square)_c$, and for the substitutional model $(B,Fe)_1$:$(\square)_c$ it is

$$^{\circ}G_{m}^{\phi} = \sum_{i=B,Fe} x_{i}\ {}^{\circ}G_{i:\square}^{\phi} + RT \sum_{i=B,Fe} x_{i} \ln x_{i} + x_{B}x_{Fe}L_{B,Fe:\square}^{\phi} + {}^{mo}G_{m}^{\phi} \quad (5)$$

In the present work, it was possible to obtain a reasonable description without using the excess energy term ${}^{xs}G_{m}^{\phi}$ for the interstitial model. The reference energy term for one mole of phase ϕ, ${}^{\circ}G_{Fe:B}^{\phi}$, in Eq. (4) is then written as

$$(c+1)\ {}^{\circ}G_{Fe:B}^{\phi} = c{}^{\circ}G_{B}^{SER} + {}^{\circ}G_{Fe}^{SER} + (c+1)(a^{\phi} + b^{\phi}T) \quad (6)$$

The interaction coefficient $L_{B,Fe:\square}^{\phi}$ in Eq. (5) can be expanded as a Redlich–Kister polynomial, similar to Eqs. (2) and (3).

In Eqs. (4) and (5), the magnetic contribution to the Gibbs energy is taken into account through the ${}^{mo}G_{m}^{\phi}$ term, as expressed by Hillert and Jarl [41]:

$${}^{mo}G_{m}^{\phi} = RT \ln(\beta^{\phi} + 1) \cdot f(\tau) \quad (7)$$

where $\tau = T/T_{C}^{\phi}$, with T_{C}^{ϕ} the Curie temperature (in K) for phase ϕ, and β^{ϕ} is the average magnetic moment per atom (in Bohr magnetons).

The Gibbs energies for one formula unit of the stoichiometric phases FeB and Fe_2B are given by:

$$^{\circ}G^{FeB} = {}^{\circ}G_{B}^{SER} + {}^{\circ}G_{Fe}^{SER} + a^{FeB} + b^{FeB}T + c^{FeB}T \ln T + \ldots \quad (8)$$

$$^{\circ}G^{Fe_2B} = {}^{\circ}G_{B}^{SER} + 2{}^{\circ}G_{Fe}^{SER} + a^{Fe_2B} + b^{Fe_2B}T + c^{Fe_2B}T \ln T + \ldots \quad (9)$$

In Eqs. (8) and (9) additional terms describing the temperature dependence are present. Only the coefficients a and b are optimized. The other coefficients are taken from the SGTE substance database [42].

Since not enough information is available on the solubility of Fe in βB [43], the terminal rhombohedral B phase is modelled as a stoichiometric phase.

4. Optimization of the model parameters

The optimization of the B–Fe system was performed using the PARROT module [44] of the Thermo-Calc software [45]. This procedure involves a weighted least-square optimization of the Gibbs energy model parameters based on experimental thermochemical and phase diagram data. In order to carry out the calculations, it is necessary to provide start values for all model parameters that are to be optimized. Proper start values are required to calculate the equilibrium state corresponding to each experimental datum.

As a first step, the model parameters for the liquid phase were obtained through a fit of the data on the partial molar heat of mixing data of B and Fe in the liquid phase [33], with the activity data in liquid alloys of Yukinobu et al. [36]. During the optimization it was impossible to obtain a suitable set of parameters that fit all experimental data. The partial molar enthalpy of mixing of B in the liquid given by Esin et al. [33] shows very negative values. Therefore, it may be suggested that these data are erroneous. Unfortunately, since the authors [33] used an approximate calculation to obtain the presented data, it is very difficult to assess the accuracy of their data. During the optimization only their data concerning the partial molar enthalpies of Fe were considered.

Next, the thermodynamic parameters for the borides FeB and Fe_2B were estimated using calorimetric measurements of the enthalpies of formation [14,21,31,35], together with invariant equilibrium data. The use of the Debye equation for the temperature dependence of the heat capacity of FeB given by Voitovich and Shakhanova [12] led to unreasonable values for the coefficients. This set of experimental data was therefore not used in the optimization.

Subsequently, the f.c.c. and b.c.c. Fe solid solutions were optimized. The activity data of B in diluted b.c.c. and f.c.c. [30] were not used in the calculations, because of the improbable values for the parameters resulting from their use. Due to lack of thermochemical data, the optimization was performed using only information on the invariant reactions (cf. Table 5), and on the phase boundaries [22].

Finally, the model parameters were further refined by simultaneous optimization of all data, including e.m.f. data [32,34]. Due to the limited accuracy of this experimental technique, these experiments were not considered during the previous steps. And a lower weight was accorded to these data during the final optimization step. This optimization strategy was used for both the interstitial as the substitutional model. The resulting values for the Gibbs energy model parameters are listed in Table 3 for the interstitial model, and in Table 4 for the substitutional model.

5. Discussion

The phase diagram calculated using the description for the interstitial model is presented in Fig. 1. In Fig. 2, the Fe-rich side of this phase diagram is compared with experimental phase boundary data. The agreement with the solubility data of Cameron and Morral [22] is very good.

Table 3
Gibbs energy parameters for the B–Fe system, according to the interstitial model presented (in SI units)

Liquid, (B,Fe)		
${}^{0}L^{L}_{B,Fe}$	=	$-130\,243+32.152\ T$
${}^{1}L^{L}_{B,Fe}$	=	$+6571$
${}^{2}L^{L}_{B,Fe}$	=	$+30\,933$
b.c.c., $(Fe)_1$:$(B,\square)_3$		
${}^{\circ}G^{b.c.c.}_{Fe:B}$	=	$3H^{SER}_{B}+H^{SER}_{Fe}-3000+12.832\ T$
$T^{b.c.c.}_{C}$	=	$1043\ x_{Fe}$
$\beta^{b.c.c.}$	=	$2.22\ x_{Fe}$
f.c.c., $(Fe)_1$:$(B,\square)_1$		
${}^{\circ}G^{f.c.c.}_{Fe:B}$	=	$H^{SER}_{B}+H^{SER}_{Fe}+45\,496-77.534\ T$
$T^{f.c.c.}_{C}$	=	$-201\ x_{Fe}$
$\beta^{f.c.c.}$	=	$-2.1\ x_{Fe}$
Fe_2B, $(Fe)_2$:$(B)_1$		
${}^{\circ}G^{Fe_2B}$	=	$H^{SER}_{B}+2H^{SER}_{Fe}-96\,363+481.992\ T$ $-79.04999\ T\ \ln T-0.007071\ T^2+731\,991\ T^{-1}$
$T^{Fe_2B}_{C}$	=	$1018\ x_B x_{Fe}$
β^{Fe_2B}	=	$1.91\ x_B x_{Fe}$
FeB, $(Fe)_1$:$(B)_1$		
${}^{\circ}G^{FeB}$	=	$H^{SER}_{B}+H^{SER}_{Fe}-93\,615+324.299\ T$ $-50.00001\ T\ \ln T-0.004999999\ T^2+530\,000\ T^{-1}$
T^{FeB}_{C}	=	$600\ x_B x_{Fe}$
β^{FeB}	=	$1.03\ x_B x_{Fe}$

As explained above, the other experimental data are assumed to be less reliable.

Table 5 compares the invariant temperatures and compositions with experimental data, and with other calculations. Both the interstitial and the substitutional model reveal an eutectoid invariant equilibrium between αFe, γFe and Fe_2B at 1184.6 K, consistent with a previous calculation [7] and with the most recent experimental

Table 4
Gibbs energy parameters for the B–Fe system, according to the substitutional model presented (in SI units)

Liquid, (B,Fe)		
${}^{0}L^{L}_{B,Fe}$	=	$-126\,220+29.44\ T$
${}^{1}L^{L}_{B,Fe}$	=	$+8390$
${}^{2}L^{L}_{B,Fe}$	=	$+33\,538$
b.c.c., $(B,Fe)_1$:$(\square)_3$		
${}^{0}L^{b.c.c.}_{B,Fe:\square}$	=	$-63\,789+55.341\ T$
$T^{b.c.c.}_{C}$	=	$1043\ x_{Fe}$
$\beta^{b.c.c.}$	=	$2.22\ x_{Fe}$
f.c.c., $(B,Fe)_1$:$(\square)_1$		
${}^{0}L^{f.c.c.}_{B,Fe:\square}$	=	$-70\,178+55.470\ T$
$T^{f.c.c.}_{C}$	=	$-201\ x_{Fe}$
$\beta^{f.c.c.}$	=	$-2.1\ x_{Fe}$
Fe_2B, $(Fe)_2$:$(B)_1$		
${}^{\circ}G^{Fe_2B}$	=	$H^{SER}_{B}+2H^{SER}_{Fe}-96\,087+481.861\ T$ $-79.04999\ T\ \ln T-0.007071\ T^2+731\,991\ T^{-1}$
$T^{Fe_2B}_{C}$	=	$1018\ x_B x_{Fe}$
β^{Fe_2B}	=	$1.91\ x_B x_{Fe}$
FeB, $(Fe)_1$:$(B)_1$		
${}^{\circ}G^{FeB}$	=	$H^{SER}_{B}+H^{SER}_{Fe}-93\,664+324.294\ T$ $-50.00001\ T\ \ln T-0.004999999\ T^2+530\,000\ T^{-1}$
T^{FeB}_{C}	=	$600\ x_B x_{Fe}$
β^{FeB}	=	$1.03\ x_B x_{Fe}$

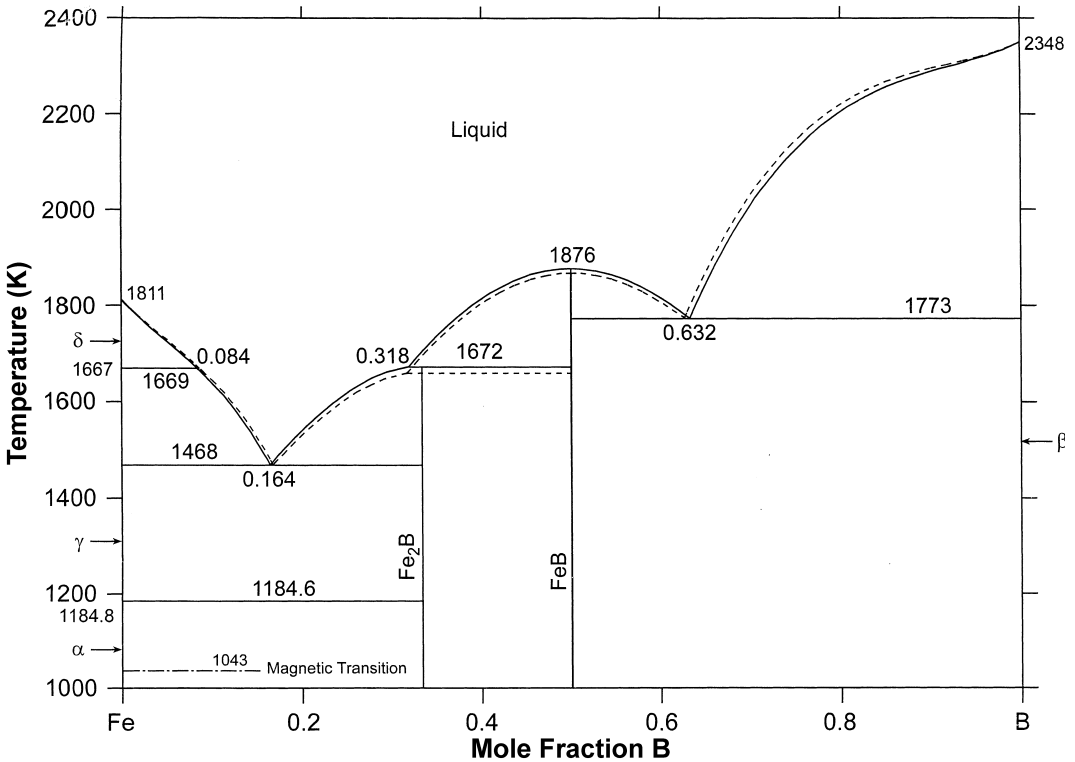

Fig. 1. Calculated B–Fe phase diagram according to the present description for the interstitial model (full lines) and for the substitutional model (dashed lines). Numerical information indicated is for the interstitial model.

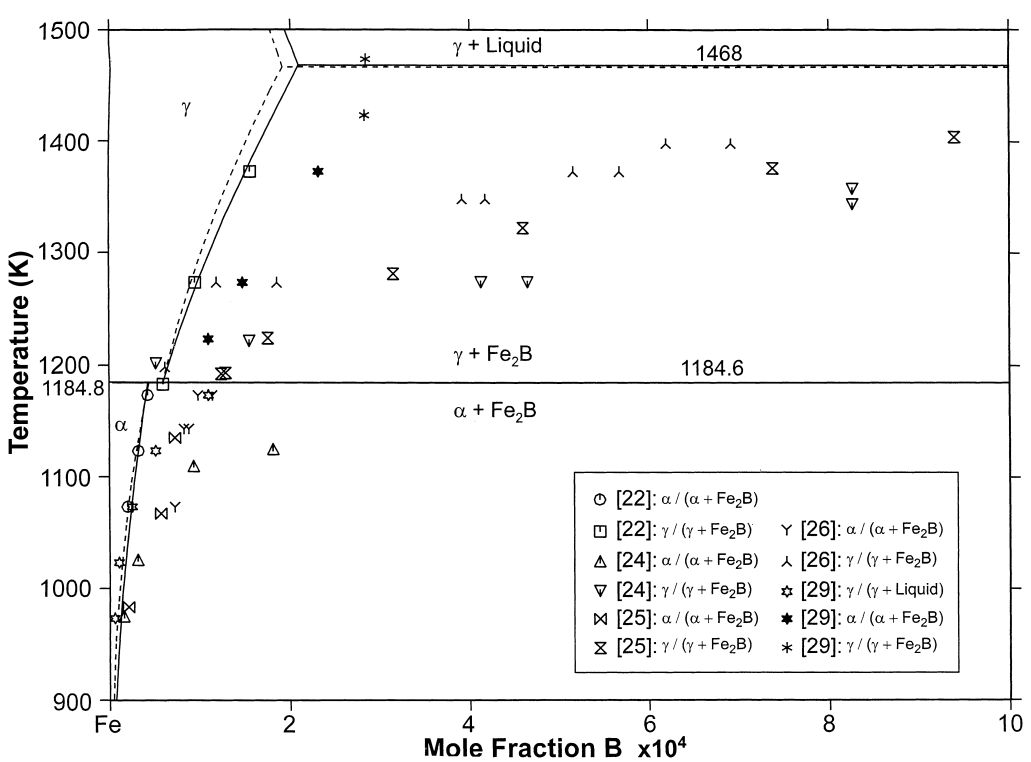

Fig. 2. Calculated Fe-rich side of the B–Fe phase diagram according to the present description for the interstitial model (full lines) and for the substitutional model (dashed lines).

Table 5
Comparison of the calculated equilibria with experimental data and with assessed and calculated values for the B–Fe system

Equilibrium	Type	T(K)		x_B		Ref.
L⇌FeB	Congruent	1876	0.5	0.5		Int.
		1866	0.5	0.5		Sub.
		1930				[4][a]
		1863				[5][a]
		1923				[6][a]
		1906				[7][b]
		1813				[19]
		1923				[20]
		1863				[21]
L⇌FeB+βB	Eutectic	1773	0.632	0.5	1	Int.
		1772	0.626	0.5	1	Sub.
		1770				[4][a]
		1770	0.64		>98.5[c]	[5][a]
		1773	0.64		>98.5[c]	[6][a]
		1782	0.633		1	[7][b]
		1701				[19]
		1773	0.64			[20]
L+FeB⇌Fe_2B	Peritectic	1672	0.318	0.5	0.333	Int.
		1658	0.316	0.5	0.333	Sub.
		1680				[4][a]
		1680	0.325			[5][a]
		1662	0.325			[6][a]
		1681	0.326			[7][b]
		1662				[19]
		1683				[20]
δFe+L⇌γFe	Peritectic[d]	1669	$4.45\cdot10^{-5}$	0.0837	$1.01\cdot10^{-4}$	Int.
		1667.9	$8.85\cdot10^{-5}$	0.0867	$1.02\cdot10^{-4}$	Sub.
		1654			0.072	[5][a]
		1662			0.072	[6][a]
		1666.9	$1.23\cdot10^{-4}$	0.081	$1.06\cdot10^{-4}$	[7][b]
		1654				[19]
L⇌γFe+Fe_2B	Eutectic	1468	0.164	$2.10\cdot10^{-4}$	0.333	Int.
		1467	0.167	$1.91\cdot10^{-4}$	0.333	Sub.
		1450				[4][a]
		1450	0.17			[5][a]
		1447	0.17			[6][a]
		1446	0.164	$1.87\cdot10^{-4}$		[7][b]
		1447	0.169			[19]
		1473				[20]
γFe⇌αFe+Fe_2B	Eutectoid[e]	1184.6	$5.95\cdot10^{-5}$	$4.28\cdot10^{-5}$	0.333	Int.
		1184.6	$5.90\cdot10^{-5}$	$4.40\cdot10^{-5}$	0.333	Sub.
		1186				[5][a]
		1184				[6][a]
		1184.6	$5.78\cdot10^{-5}$	$4.52\cdot10^{-5}$		[7][b]
		1188				[19]
		1183	$6.00\cdot10^{-5}$	$4.25\cdot10^{-5}$		[22]

Values for both models described in this work were given: interstitial (Int.) and substitutional (Sub.) model.
[a] Assessed value (not used in the optimization).
[b] Calculated value (not used in the optimization).
[c] The solubility of Fe in βB is not being modelled. This value is therefore not used.
[d] According to [5–7,19] this reaction is of the metatectic type.
[e] According to [5,6,19] this reaction is of the peritectoid type.

information [22,30]. In spite of the fact that a metatectic invariant reaction between L, δFe and γFe was assumed by different assessments [6,15], a peritectic type of reaction followed from the calculations in this work. It should however be noticed that due to the extremely low solubility of B in the Fe solid solutions these invariant equilibria are very close to being degenerated.

In Fig. 3 the partial molar enthalpies of mixing in the liquid at 1950 K, calculated with the interstitial model, are compared with experimental data [33] and with other calculations [7,8,10]. For Fe, the results seem to correspond very well, whereas for B there is a discrepancy. As mentioned before, the experimental values for B by Esin et al. [33] might be inaccurate. The calculated temperature

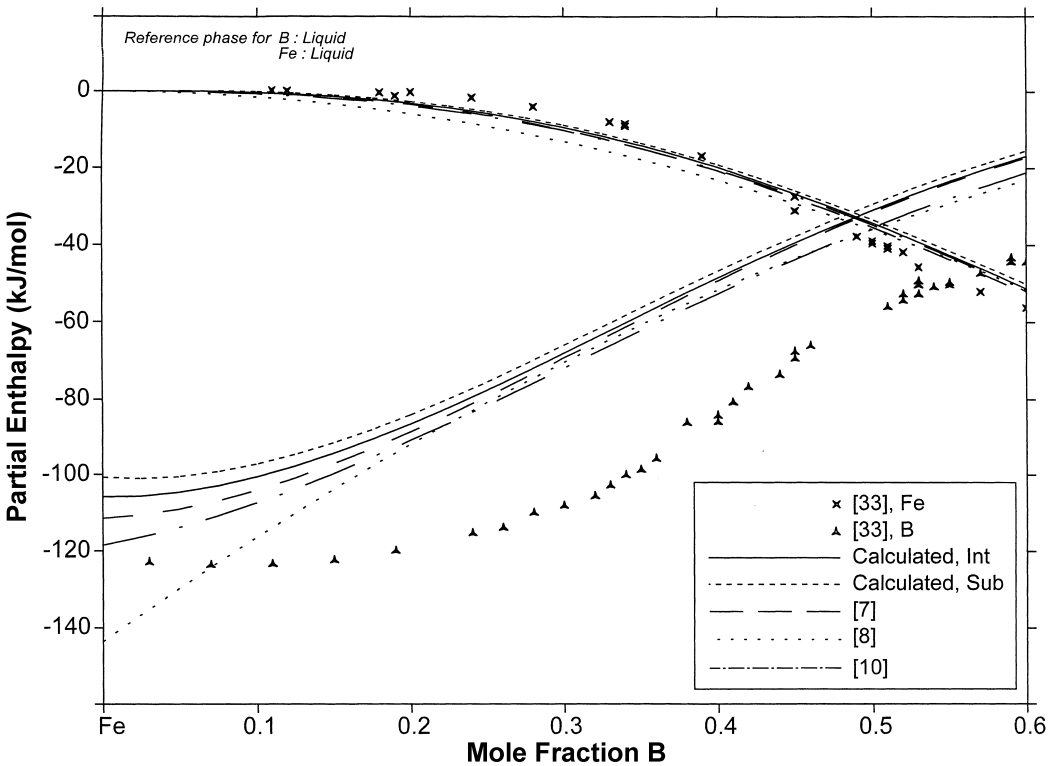

Fig. 3. Partial enthalpy of mixing in liquid B–Fe alloys at 1950 K.

dependence of the enthalpy of formation of the monoboride FeB according to the interstitial model is given in Fig. 4. The agreement with experimental data [14,21] is very good. Experimental values for the activity of B in the liquid phase at 1673 K [36] are compared with the calculations for the interstitial model in Fig. 5. The

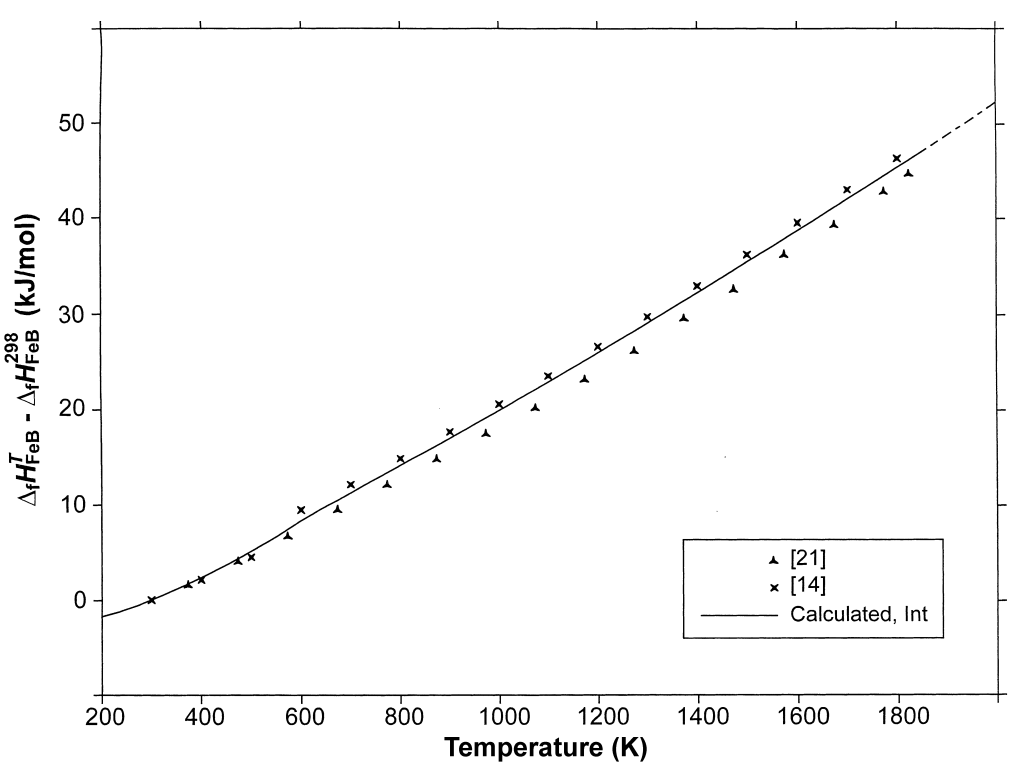

Fig. 4. Enthalpy of formation of FeB as a function of temperature, relative to the enthalpy of formation at 298 K.

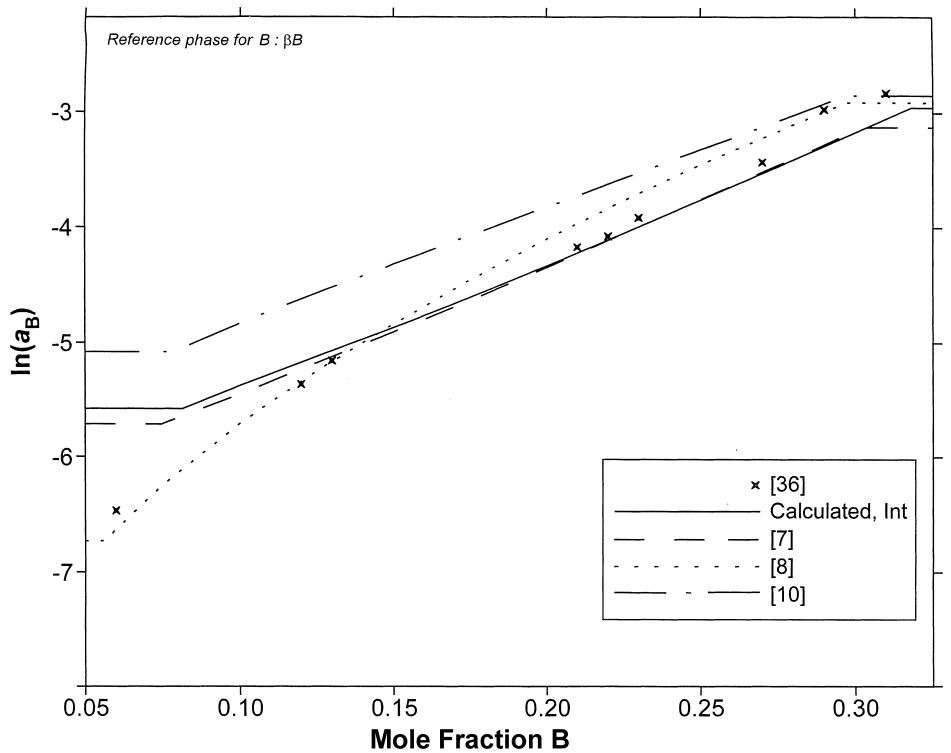

Fig. 5. Activity of B in liquid B–Fe alloys at 1673 K.

accordance between both is quite good. Other calculated results [7,8,10] are added for comparison.

6. Conclusion

Two optimized descriptions for the B–Fe system were developed, considering B as an interstitial species, or as a substitutional species in the three Fe solid solutions. The optimization was done using the PARROT module of the Thermo-Calc software. Both descriptions presented were in good agreement with experimental data and can be used for extrapolation to higher-order systems.

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